Do Storms Bring Crime?

Evidence from US Counties

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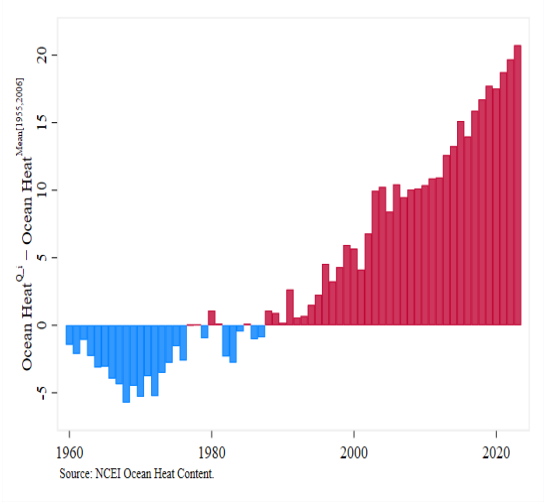
Job Market Paper

Warmer Oceans & Stronger Storms



The New York Times

Record Hot Water Is Fueling Hurricane Milton

Milton grew into a Category 5 hurricane in less than a day as it crossed warm oceans across the Gulf of Mexico.

By Austyn Gaffney and Mira Rojanasakul

Oct. 8, 2024

Hurricane Milton in the Gulf of Mexico on Tuesday morning. NOAA, via Agence France-Presse — Getty Images

Motivation

- Hurricanes are among the most frequently occurring, costly natural disasters in the US.
- From 1980–2020, **78** landfalling hurricanes caused **$1.3T** in damages ≈ **4.4%** of US GDP. (Muller et al. 2025) Distribution
- But hurricane effects are not only limited to loss of life and property.
- Hurricanes can cause substantive socioeconomic disruptions.
  - In the short-run: **Increased** out-migration often selectively displacing the economically better-off. (Anand et al. 2024; Gellman et al. 2024; Hsiang and Jina 2014)
  - In the long-run: **Falling** incomes and employment rates. (Jerch et al. 2023; Deryugina 2017; Anttila-Hughes and Hsiang 2013)
- Such socioeconomic changes create opportunities for crime.
- Especially property crimes. (Huang 2022; Augusto 2021; Frailing and Harper 2020; Spencer 2017)

This Paper

- **Question:** How do hurricanes of varying intensities affect index crimes?
- **Key Outcomes:** County-level crime and arrest rates (per 10,000), 1980–2020.
- **Setting:** All landfalling Atlantic hurricanes, 1990-2010.
- **Treatment:** County-level exposure to maximum sustained wind speed experienced during a storm.
  - **Minor** $\rightarrow$ Wind speed $\in [64, 95)$ knots
  - **Major** $\rightarrow$ Wind speed $\in [95, 123.3)$ knots

  } Saffir Simpson Scale
- **Empirical Strategy:** Multiple event-study specification. (Sandler and Sandler 2014)
  - Allows for multiple/overlapping events per unit, having multiple event dummies $\neq 0$ simultaneously.
  - **Lesser** bias than first events or stacked event-study.
  - Officer peer death (Cho et al. 2024); Officer peer injury (Holz et al. 2023); E-cigarette tax change (Allcott and Rafkin 2022); Homestead exemptions (Pattison and Hynes 2020)

Preview of Key Findings

- **Main Results**
  - Minor hurricanes show **little** to no crime impact.
  - Effects of major hurricanes are **mostly centered** around property crimes.
  - **Major hurricanes** cause property crimes to $\uparrow$ **8.5% ($\equiv$ 13%)** $\Rightarrow$ **\$385,821**, in the decade after.
  - **Strongest** increases occur in incidents of vehicle thefts.
  - Treated counties, **without** prior (1970-89) hurricane exposure, see **2.5×** more property crimes.
- **Heterogeneity Results**
  - Socially **vulnerable** counties, experience notably **larger** effects.
  - Counties with **smaller** black population share, witness **4×** more property crimes.
  - While estimated arrest rate for black offenders is **3.5×** non-black offenders.
  - Compared to hurricanes, other high-wind disasters have **little** effect on index crimes.

Contribution & Relevant Literature

- Estimate **short** (1-4 years) and **long-run** (5-10 years) effects of county-level hurricane exposure for **all** landfalling hurricanes between 1990-2010.
  - Single event (Zhao and Zhang 2025; Augusto 2021; Linning and Silver 2021; Frailing et al. 2021; Leitner et al. 2011; Leitner and Helbich 2011); Short treatment window (Spencer and Strobl 2019; Spencer 2017) ; Short panel (Huang 2022)
- **New** evidence on the role of prior hurricane exposure.
  - Larger GDP losses among countries with little prior cyclone exposure. (Hsiang and Jina 2014)
- Discussion on potential mechanism – **evacuation** orders and **outmigration**.
  - Property crimes increase post evacuation owing to reduced guardianship (Linning and Silver 2021; Leitner and Helbich 2011); Greater inclination of evacuation among the economically better-off (Gellman et al. 2024; Strobl 2011)
- Heterogeneity analysis, varying by – Socioeconomic characteristics (bottom vs. top quartile); Offender race (black vs. white); Disaster types (hurricanes vs. other high wind disasters).
  - Storm warnings (Spencer and Strobl 2019); Socioeconomic status (Huang 2022)

Data I

- **Period:** Balanced panel of annual county-level outcomes, 1980–2020.
- **Crime:** FBI's Uniform Crime Reports (UCRs)
  - Monthly agency-level data on reported crimes and arrests (by age, sex, and race).
  - Focus on **index crimes**: $\underbrace{\text{Murder, Rape, Assault, Robbery}}_{\text{Violent}}$ $\underbrace{\text{Burglary, Theft, Vehicle Theft}}_{\text{Property}}$
    - Reporting to UCR is voluntary, and not all agencies report consistently.
    - Agency-month data is carefully cleaned and aggregated to the county-year level. Cleaning
- **Hurricane:** Best Track Data from the Atlantic Hurricane Database (HURDAT2)
  - Wind-speed data for storm events recorded at a six hour interval at the lat-long level.
  - Following (Anderson et al. 2020), create county-level wind exposure for all landfalling hurricanes.
    - For each storm (1990–2010), record the **maximum sustained wind** speed experienced by a county.
    - Counties experiencing multiple storms a year, retain only the **highest** wind-speed observation.
    - Resulting in **unique** storm-year-wind speed observation, for each county.

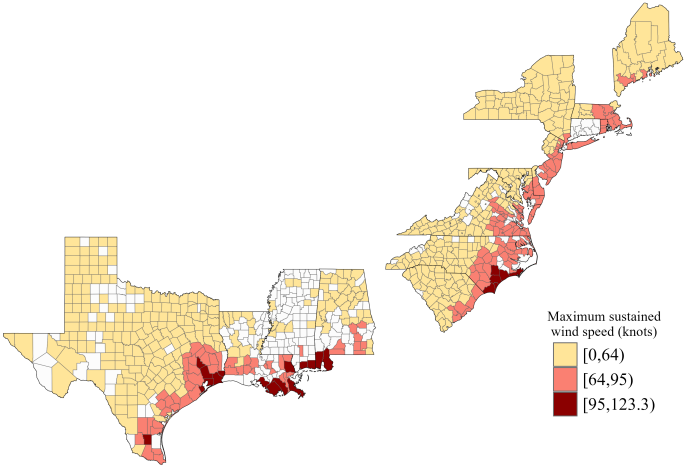
Data II

- **Controls:**
  - Baseline (1980) Characteristics: National Historical Geographic Information System (NHGIS)
  - Demographics: Surveillance, Epidemiology, and End Results (SEER)
- **Heterogeneity:**
  - Other Disasters: NOAA's Storm Events Database (SED)
    - Using **property damage estimates**, classify other high-wind disasters as **minor** or **major** events.
- **Mechanism:**
  - Economic: Quarterly Census of Employment & Wages (QCEW) + Bureau of Economic Analysis (BEA)
    - Use data on **employment** from QCEW and **per-capita incomes** from BEA.
  - Hurricane Evacuations (2014-2022): (Anand et al. 2024)
    - Data on hurricane evacuation order types (**voluntary/mandatory**) linked to HURDAT2 information.
    - Reports evacuation order types (voluntary/mandatory) for 25 hurricanes (2014–2022)
    - Linking with my hurricane data gives 435 **unique** storm-wind speed-evacuation type observations.
    - Evacuation data is matched with hurricane information at the county level
    - Resulting in 435 **unique** observations defined by storm, wind speed, and evacuation type
  - Migration (1983-2020): IRS, County to County Flows
    - I focus on **net migration flows**.

Sample

- Starting with all Atlantic coast states.
- Cleaning the crime data and imposing a balanced panel requirement **excludes** Florida and 319 other counties.
- Restrict sample to states impacted by **at least one** hurricane between 1990–2010.
- 503 **control** counties.
- 186 **treated** counties.

Cleaning Crime Data



Summary Statistics

- Pre-treatment balance is hard.
- Level differences exist between control and treatment.
- Might or might not lead to differences in trends.
- But, if it does:
  - Control for: $\mathbf{X}'_{c,1980}\mu_t$
  - Robustness to alternate control groups.
  - Event-study to visually inspect pre-trends.

Table 1: Baseline (1980) Characteristics

	Hurricane counties	Non-hurricane counties	*p*-value
Distance to nearest coastline *(miles)*	27	187	<0.001
Land area *(square miles)*	587.79	770.56	<0.001
Property Crime *(per 10k)*	376.05	285.76	0.001
Violent Crime *(per 10k)*	42.77	26.67	0.007
Police Officers *(per 10k)*	186.14	150.72	0.003
Population	174,859.49	74,800.33	<0.001
Population density *(per square mile)*	938.81	229.65	0.046
Black	23.11%	12.97%	<0.001
Male	48.85%	48.66%	0.234
No high school *(aged $\geq$ 25 years)*	42.49%	43.57%	0.255
Annual average wages *(in US$)*	12,283.04	11,719.74	0.008
Poverty rate	16.49%	15.93%	0.32
Unemployment rate	6.26%	5.68%	0.003
Number of Counties	186	503	

Descriptive Trends

Figure 1: Minor Hurricane

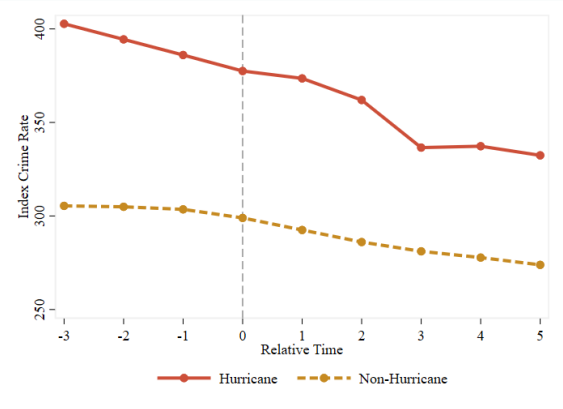
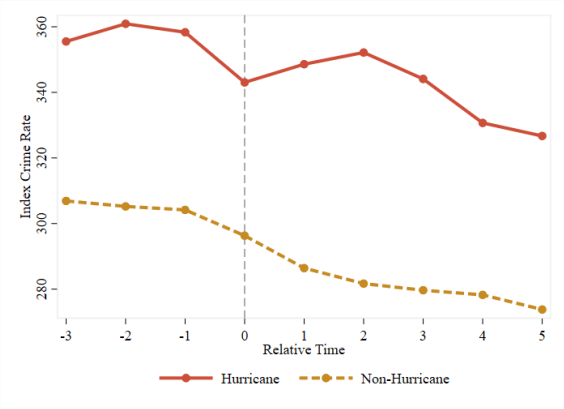


Figure 2: Major Hurricane



Specification

- **Event Study**

$$Y_{ct} = \sum_{\tau=-10,\tau\neq-2}^{10} \beta_\tau Min_{c\tau} + \sum_{\tau=-10,\tau\neq-2}^{10} \kappa_\tau Maj_{c\tau} + \mu_c + \mu_t + \mathbf{X}'_{c,1980}\mu_t + \epsilon_{ct} \quad (1)$$

  - $Min_{ct}/Maj_{ct} = \begin{cases} 1; \text{ if county } c \text{ as of year } t \text{ experienced a minor (major) hurricane } \tau \text{ years ago} \\ 0; \text{ otherwise} \end{cases}$
  - β_τ and κ_τ: Dynamic effects of minor and major hurricanes.
  - μ_c and μ_t: County and year FEs.
  - $\mathbf{X}'_{c,1980}$: Baseline (1980) characteristics interacted with year FE.
  - SEs clustered at county-level and hurricane indicators combined into two-year bins, except event year.

- **Pooled**

$$Y_{ct} = \beta_1 Min_{ct}^{1-4} + \beta_2 Min_{ct}^{5-10} + \kappa_1 Maj_{ct}^{1-4} + \kappa_2 Maj_{ct}^{5-10} + \mu_c + \mu_t + \mathbf{X}'_{c,1980}\mu_t + \epsilon_{ct} \quad (2)$$

  - Omit lead indicators, pool post-hurricane indicators into two bins: 1-4 years & 5-10 years after.
  - β_1/κ_1 & β_2/κ_2: Short- & long-term effects of minor (major) hurricanes in the 1-4 & 5-10 years after.

- **Heterogeneity**

$$Y_{ct} = \beta Min_{ct}^{1-10} + \kappa Maj_{ct}^{1-10} + \mu_c + \mu_t + \mathbf{X}'_{c,1980}\mu_t + \epsilon_{ct} \quad (3)$$

Empirical Strategy

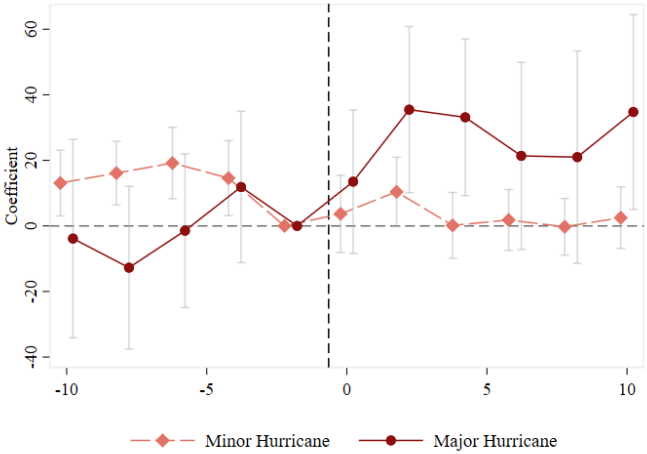
- Use multiple event study, to identify hurricane effects on crime. (Sandler and Sandler 2014)
- To be able to fully identify all ten leads and lags for each hurricane event, I consider storms, only occurring between 1990-2010. (Deryugina 2017)
- This also allows to define a **pre-treatment baseline** period, 1980.
- Instead of time-varying controls, I use year indicators interacted with 1980 county characteristics[1], to reduce possibility of pre-trends. (Hoynes and Schanzenbach 2009; Deryugina 2017)
- Identifying variation comes from **never-treated** counties, with alternate control groups for robustness. Robustness Estimate
  - Geographic controls; Matched controls; Not-yet treated counties; Last treated cohort.
- **Identifying Assumption**: Conditional on fixed effects, hurricane counties would have followed the same trend in crime rates as non-hurricane counties.

[1] Per capita law enforcement, index crime rate, log of population density, log of distance from coast, % black, % male, % poor, % unemployed and % aged $\geq$ 25 without HS degree.

Event Study

- ∥ trends for major but pre-trends for minor.
  - Few exclusively major counties.
  - Sig. differences at baseline.
  - Lack of good comparison.
- Remedies:
  - Using only treated counties.
  - (Sun and Abraham 2021)
- Effects centered around property crimes.
- No long-term impact of minor storms.
- Major hurricane cause sustained increases in (property) crimes in the decade after.
- Larger effects in the short-run.

Figure 3: Index Crime Rate



Impact of Hurricanes on Index Crimes

- Results from Pooled specification (Eq-2).
- Use (Autor et al. 2017) per-crime cost and adjust for 2025 prices. (Mello 2019)
- Negligible crime effect of minor hurricanes.
- No noticeable effect on violent crimes.
- Relative to the baseline, **major hurricanes** cause **property crimes** in the:
  - SR: ↑ by 9.1% (≡ 14.3%) ⇒ ↑$ 209,743
  - LR: ↑ by 7.6% (≡ 10.7%) ⇒ ↑$ 176,078

	Property Crime Rate	Violent Crime Rate
Panel A: *Effect of Minor Hurricanes*		
Minor Hurricane in last 1–4 years	-1.21	-0.89
	(4.17)	(0.97)
Minor Hurricane in last 5–10 years	-6.31	-0.49
	(4.25)	(1.03)
Panel B: *Effect of Major Hurricanes*		
Major Hurricane in last 1–4 years	**34.36*** **	1.49
	(11.66)	(3.00)
Major Hurricane in last 5–10 years	**28.84** **	1.86
	(11.97)	(2.68)
Baseline Mean of Dep. Var	376.05	42.77
Victimization Cost	$6,105	$101,827
Observations	28,249	28,249

Impact on Individual Index Crimes

- Results from (Eq-2).
- $Hurr^{Maj} \rightarrow$ sig.$\uparrow$ across **all** property crimes.
- Largest spike in MV Thefts: $\uparrow$ 25.6% ($\equiv$ 27.7%)
- $Hurr^{Min} \not\rightarrow$ property crimes.
- Negligible effect of either hurricane type on most violent crimes, except:
  - Rape: $Hurr^{Min} \uparrow$; $Hurr^{Maj} \downarrow$

Event Plot

	Larceny Theft	Burglary	Vehicle Theft	Murder	Rape	Robbery	Assault
Panel A: *Effect of Minor Hurricanes*							
Minor Hurricane in the last 1–4 years	-1.47	0.83	-0.58	0.05*	0.17**	0.50	-1.60**
	(2.99)	(1.29)	(0.72)	(0.03)	(0.07)	(0.42)	(0.76)
Major Hurricane in the last 5–10 years	-3.67	-1.62	-1.01	-0.01	0.10	0.40	-0.99
	(3.02)	(1.40)	(0.82)	(0.02)	(0.11)	(0.46)	(0.82)
Panel B: *Effect of Major Hurricanes*							
Major Hurricane in the last 1–4 years	16.79**	9.86**	7.71***	-0.02	-0.37**	2.06*	-0.19
	(7.08)	(4.94)	(1.85)	(0.05)	(0.18)	(1.20)	(2.53)
Major Hurricane in the last 5–10 years	15.95**	5.53	7.36***	-0.02	0.15	2.09	-0.35
	(7.67)	(5.15)	(2.01)	(0.04)	(0.30)	(1.38)	(1.68)
Baseline Mean of Dep. Var	218.29	126.9	30.86	0.93	2.52	13.63	25.69
Observations	28,249	28,249	28,249	28,249	28,249	28,249	28,249

Event Plots

Does prior hurricane exposure drive crime?

Figure 4: Property Crime Rate

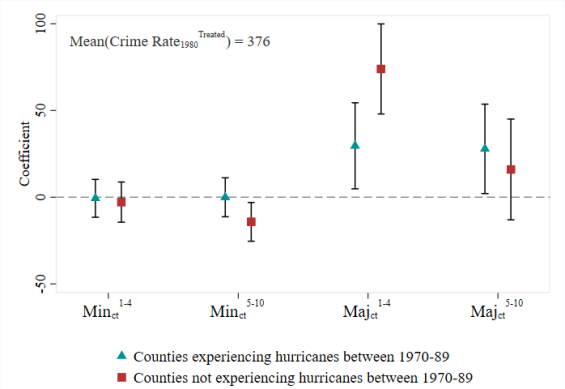
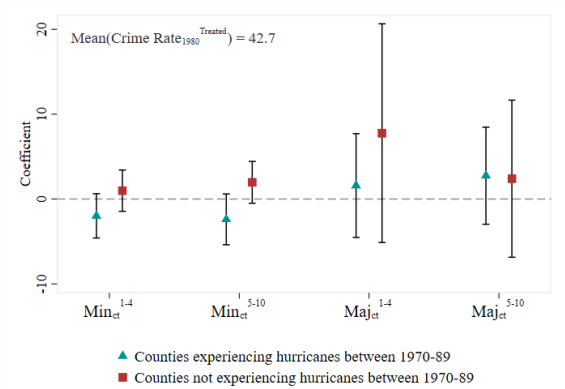


Figure 5: Violent Crime Rate



- In more **ill-prepared** counties, ones **without prior hurricane** exposure : Table-1 Table-2 Exposure Map
  - Major hurricanes cause **larger** increases in both property and violent crimes.
  - While **significantly** affecting property crimes in the SR: ↑20% ≡ (27%) ⇒ ↑$451,464
  - Effects that are **2.5×** larger.

Short-Run Mechanisms: Evacuation Orders & Outmigration

Figure 6: Evacuation Order & Hurricane Types

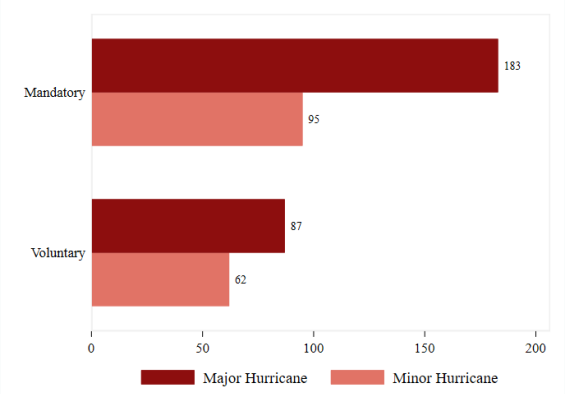
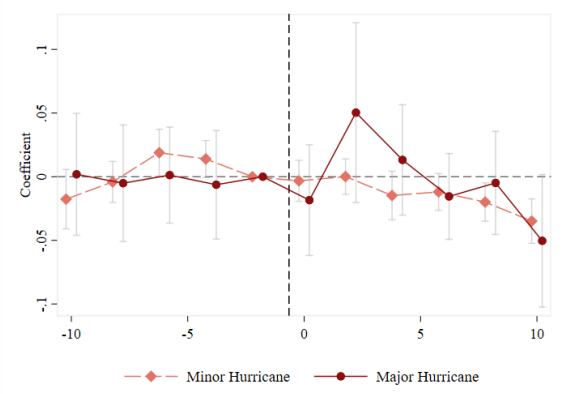


Figure 7: Log (Net Outflow)



- $Hurr^{Maj} \approx 2\times\ Evac^{Mandatory}$ + Non-binding $Evac^{Voluntary}$ (Gellman et al. 2024) $\rightarrow$ $Hurr^{Maj}$ $\uparrow$ Net-Outflow by 5%
- Increased out-migration $\rightarrow$ $\downarrow$ **SS** of human victims & $\uparrow$ **SS** of abandoned property.

Long-Run Mechanisms: Falling Incomes & Employment Rates

Figure 8: Employment Rate (15-65)

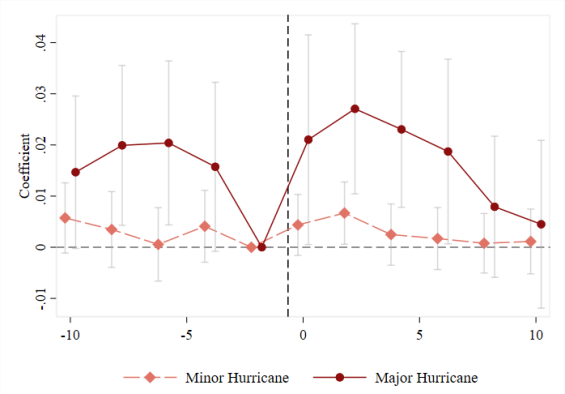
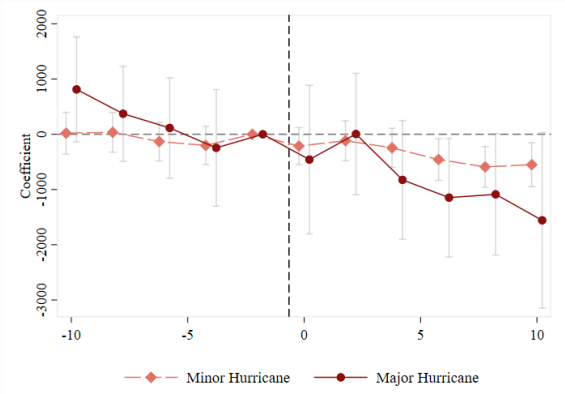


Figure 9: Per-Capita Income



- In the long-term: Employment rates $\approx \downarrow$ **2 pp** ; Per-capita incomes $\approx \downarrow$ **$1500**
- **Decreased** opportunity cost of crime. (Becker 1968)

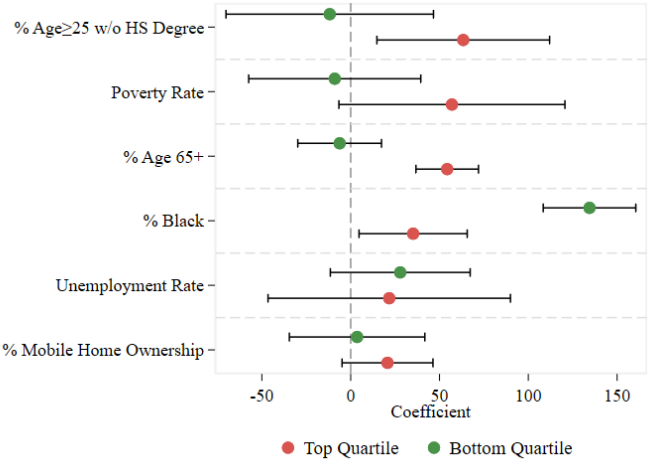
Robustness Checks

1. Sensitivity to different control groups. Table
2. Using first events only to re-estimate the main specification. Table
3. Using (Sun and Abraham 2021), to address concerns with TWFE estimator. Table
4. Controlling for multiple hypothesis testing. Table
5. Sensitivity to different sample constructions. Table
6. Sensitivity to excluding: Table
   - Extreme hurricane years of hurricanes Katrina and Rita.
   - Year of COVID-19.
   - Counties reporting extreme crime values.
   - Counties reporting less than three months of crime data.

Heterogeneity: Socioeconomic Characteristics

- Select **six** county characteristics, measured as of 1980 (pre-treatment baseline).
- For each baseline trait, sample split into **top** and **bottom** quartile, Eq-3 is ran.
- Showing results for major hurricanes.
- Notable heterogeneity exists, across traits.
- **Near-zero effects** on index crimes for counties in the **bottom quartile** of:
  (a) Share of old,
  (b) Poverty rate,
  (c) Low education,
  (d) Share of mobile homes.
- Counties in the **top quartile** of Black residents see 4× **fewer** index crimes.

Figure 10: Index Crime Rate

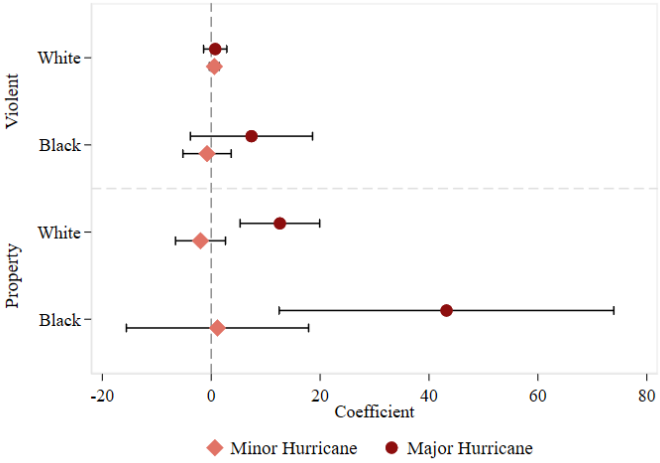


Heterogeneity: Racial Decomposition of Arrested Offender

- Examine hurricane effects on race-wise proportionate arrest rate,i.e $\Rightarrow$
  Black Arrest Rate $= \frac{\text{Black Arrests}}{\text{Black Population}} \times 10,000$
- Using heterogeneity spec. (Eq-3).
- Negligible effect of minor hurricanes across both crime and race types.
- Compared to baseline, major hurricanes:

	White	Black	Effect Size (Black/White)
Violent	↑6%	↑15.5%	10×
Property	↑26%*	↑27%*	3.5×

Figure 11: Arrest Rates by Race Type



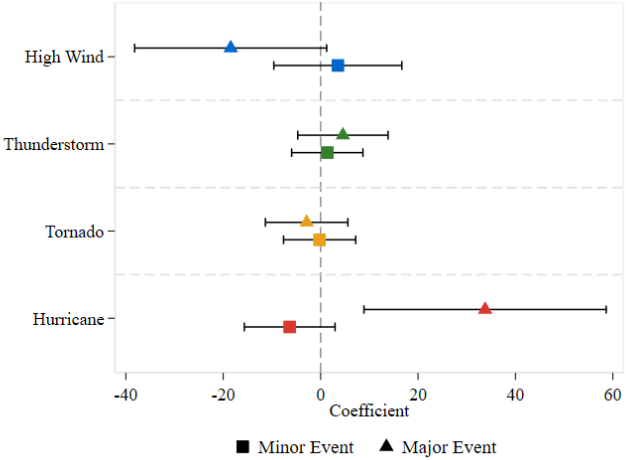
Event Plot

Heterogeneity: Other High Wind Disasters

- NOAA's SED, provides **property damage** estimates across disaster types.
- Use **median** damage to classify disasters: Below → Minor ; Above → Major.
- Define minor (major) hurricanes as before.
- Each disaster, estimate effects using Eq-3.
- Only **major** hurricanes sig. affect crime.
- Though, tornadoes are almost **as costly**!
  - Tornadoes **do not cause**:
    - Short-term ↑ Outmigration
    - Long-term ↑ Incomes

Tornado Event Plot

Figure 12: Index Crime Rate



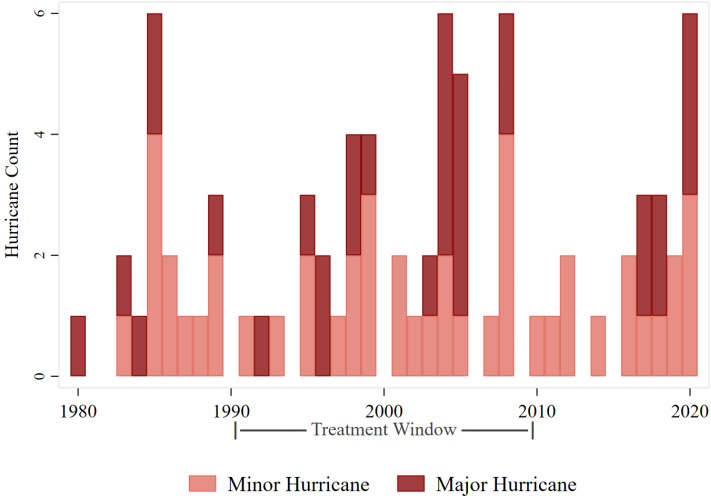
Conclusion

- I examine the effect of all landfalling Atlantic hurricanes (1990-2010), on county-level crime rates.
- While major hurricanes caused property crimes to significantly $\uparrow$ by 8.5%, in the decade after.
- Minor hurricanes had a negligible effect.
- Present heterogeneous effects of hurricanes, along a number of different dimensions.
- Analyze potential mechanisms:
  - Short-Run: $Hurr^{Major} \rightarrow \uparrow$ Out-migration $\rightarrow \uparrow$ Opportunities for property crime
  - Long-Run: $Hurr^{Major} \rightarrow \downarrow$ Per-capita Incomes and Employment Rate $\rightarrow \downarrow$ OC for crime.
- **Takeaways**
  - Even in developed countries, **vulnerable** or **ill-prepared** communities bear greater social costs of natural disasters.
  - Stronger hurricanes from global warming will further increase these costs.
  - Governments must thus equip vulnerable communities with targeted preparedness and resilience measures.

Thank you!

Appendix

Hurricane Distribution



Back

Crime Data Cleaning

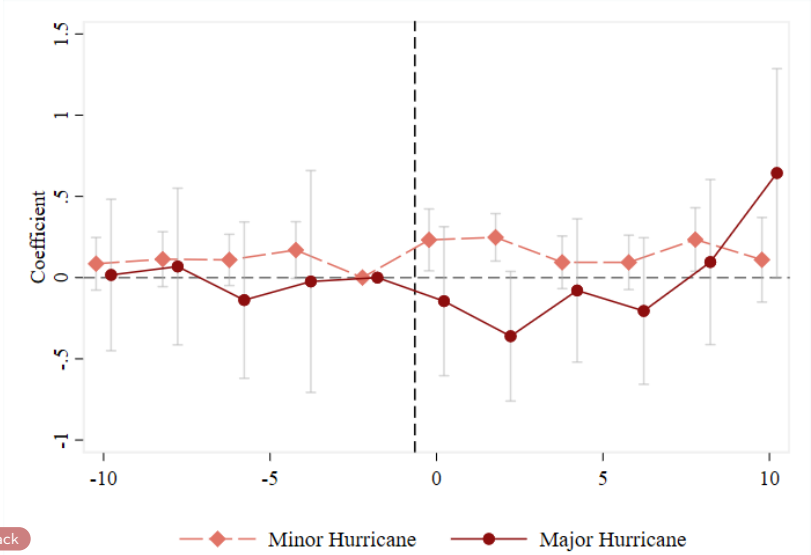
- Using (Kaplan 2024)'s concatenated Return-A files (monthly agency-level index crimes), focus on Atlantic coast states.
- To address inconsistent monthly reporting, I aggregate the crime data to the agency-year level (1,272 counties; 12,160 agencies).
- Sample restrictions:

1. Exclude agencies with negative crime values (162 removed).
2. Drop agencies with populations <1,000 and which ever fail to report crime data (3,302 removed). (Mello 2019)
3. Exclude all tribal agencies (26 removed). (Mello 2019)

- These sample restrictions leaves me with 8,105 agencies in 1,261 counties
- After which I enforce the balanced panel requirement, reducing the sample to 890 counties.

◀ Back to Data ◀ Back to Sample

Event Study: Rape



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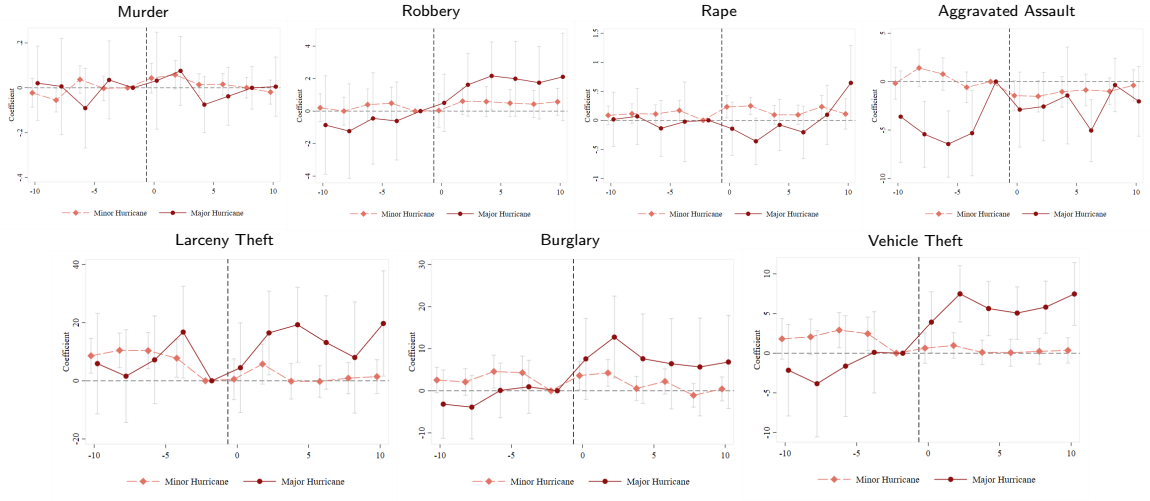
Event Study: Individual Index Crimes



Back

Does prior hurricane exposure influence crime outcomes?

	Property Crime Rate	Violent Crime Rate
Panel A: *Counties experiencing hurricanes between 1970-89*		
Minor Hurricane in last 1–4 years	0.38	-2.21*
	(5.60)	(1.34)
Minor Hurricane in last 5–10 years	1.33	-2.64*
	(5.71)	(1.53)
Major Hurricane in last 1–4 years	30.02**	1.46
	(12.76)	(3.11)
Major Hurricane in last 5–10 years	28.24**	2.71
	(13.29)	(2.91)
Baseline Mean	390.7	50.3
Panel B: *Counties not experiencing hurricanes between 1970-89*		
Minor Hurricane in last 1–4 years	-3.17	1.28
	(5.94)	(1.25)
Minor Hurricane in last 5–10 years	-14.97***	2.38*
	(5.72)	(1.27)
Major Hurricane in last 1–4 years	72.81***	8.29
	(12.79)	(6.42)
Major Hurricane in last 5–10 years	14.27	2.58
	(13.76)	(4.63)
Baseline Mean	361.67	35.37
Observations	28,249	28,249

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Does prior hurricane exposure influence crime outcomes?

	Larceny Theft	Burglary	Vehicle Theft	Murder	Rape	Robbery	Assault
Panel A: *Counties experiencing hurricanes between 1970-89*							
Minor Hurricane in last 1–4 years	1.06	0.19	-0.87	0.06*	0.10	-0.14	-2.23**
	(4.02)	(1.67)	(0.95)	(0.03)	(0.08)	(0.57)	(1.07)
Minor Hurricane in last 5–10 years	5.77	-2.94	-1.50	-0.01	0.07	-0.20	-2.51**
	(3.88)	(1.79)	(1.23)	(0.03)	(0.13)	(0.66)	(1.23)
Major Hurricane in last 1–4 years	13.22*	9.39*	7.42***	-0.04	-0.32*	1.92	-0.10
	(7.72)	(5.18)	(1.90)	(0.06)	(0.19)	(1.20)	(2.66)
Major Hurricane in last 5–10 years	14.33*	6.44	7.47***	-0.01	-0.00	2.14	0.57
	(8.69)	(5.54)	(2.15)	(0.04)	(0.26)	(1.43)	(1.86)
Baseline Mean of Dep Var.	216.2	136.5	38	0.9	2.6	17.5	29.2
Panel B: *Counties not experiencing hurricanes between 1970-89*							
Minor Hurricane in last 1–4 years	-4.81	1.64	-0.00	0.03	0.21*	1.19**	-0.15
	(4.03)	(2.01)	(0.99)	(0.04)	(0.12)	(0.57)	(0.95)
Minor Hurricane in last 5–10 years	-14.78***	0.13	-0.32	-0.01	0.11	0.96	1.32
	(3.91)	(2.13)	(1.07)	(0.04)	(0.13)	(0.68)	(0.86)
Major Hurricane in last 1–4 years	42.57***	17.54***	12.71***	0.17***	-0.75	5.02**	3.84
	(12.10)	(2.80)	(2.78)	(0.06)	(0.48)	(2.46)	(7.93)
Major Hurricane in last 5–10 years	3.26	2.40	8.61***	-0.16**	1.95	4.24	-3.45
	(8.07)	(5.20)	(2.47)	(0.07)	(1.58)	(3.04)	(5.09)
Baseline Mean of Dep Var.	220.25	117.5	23.9	0.95	2.4	9.76	22.24
Observations	28,249	28,249	28,249	28,249	28,249	28,249	28,249

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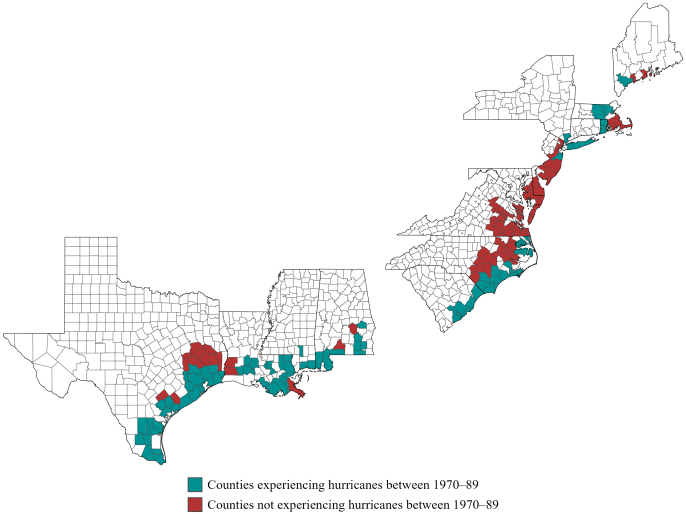
Prior Hurricane Exposure Map



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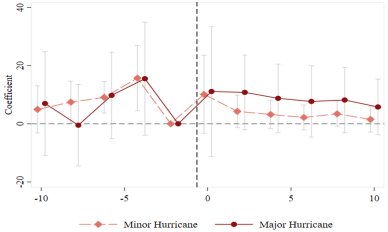
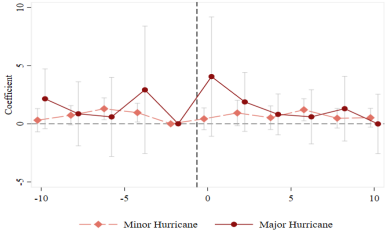
Event Study: Race wise Arrests

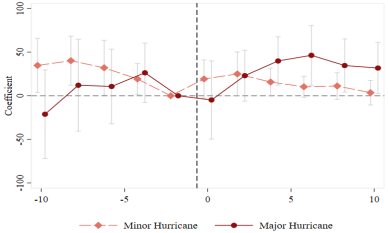
Arrest Rate for Violent Crimes by Black



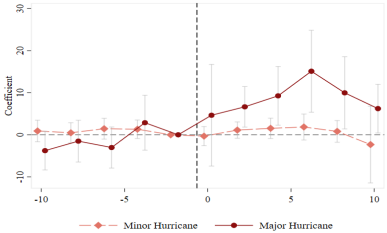
Arrest Rate for Violent Crimes by White



Arrest Rate for Property Crimes by Black



Arrest Rate for Property Crimes by White



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Event Study: Tornado

Figure 13: Log (Net Outflow)

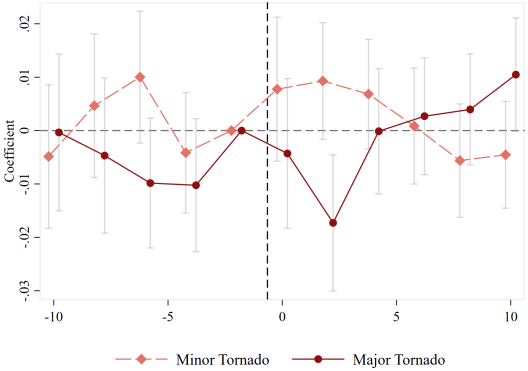
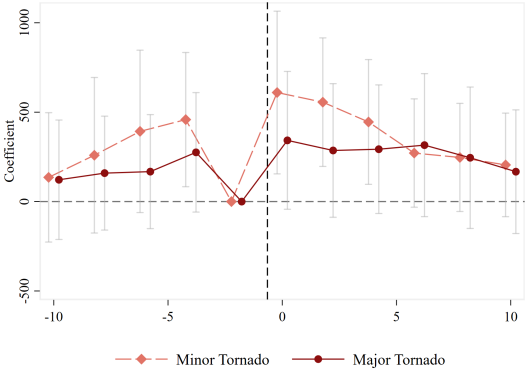


Figure 14: Per-capita Income



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Sensitivity to the choice of different control groups

	(1)	(2)	(3)	(4)
Dependent Variable: *Property Crime Rate*				
Minor Hurricane in last 1–4 years	-1.214	1.990	0.00377	-3.521
	(4.17)	(4.38)	(4.28)	(4.56)
Minor Hurricane in last 5–10 years	-6.306	-3.317	-4.629	-3.236
	(4.25)	(4.67)	(4.45)	(4.90)
Major Hurricane in last 1–4 years	34.36***	36.18***	33.51***	31.79***
	(11.66)	(11.67)	(11.69)	(11.42)
Major Hurricane in last 5–10 years	28.84**	26.56**	25.00**	23.15*
	(11.97)	(12.69)	(12.42)	(12.37)
Baseline Mean of Dep Var.	376.05	376.05	376.05	376.05
Dependent Variable: *Violent Crime Rate*				
Minor Hurricane in last 1–4 years	-0.888	-0.822	-0.944	-1.052
	(0.97)	(1.09)	(1.07)	(1.18)
Minor Hurricane in last 5–10 years	-0.489	-0.369	-0.267	-0.654
	(1.03)	(1.19)	(1.13)	(1.24)
Major Hurricane in last 1–4 years	1.489	2.708	2.534	1.507
	(3.00)	(3.17)	(3.21)	(3.28)
Major Hurricane in last 5–10 years	1.864	2.556	2.724	1.207
	(2.68)	(2.82)	(2.81)	(3.00)
Baseline Mean of Dep Var.	42.77	42.77	42.77	42.77
Observations	28249	12177	14432	7626

Note: Note: Col (1): Baseline sample. Col (2): Geographic controls. Col (3): Matched controls. Col (4): Treated only counties.

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Sensitivity to the choice of different specifications

	(1)	(2)	(3)	(4)
Dependent Variable: *Property Crime Rate*				
Minor Hurricane in last 1–4 years	-1.214	-0.509	0.246	0.246
	(4.17)	(4.23)	(4.25)	(4.25)
Minor Hurricane in last 5–10 years	-6.306	-3.626	-2.999	-2.999
	(4.25)	(4.53)	(4.49)	(4.49)
Major Hurricane in last 1–4 years	34.36***	29.79**	29.55**	29.78*
	(11.66)	(13.66)	(13.69)	(17.49)
Major Hurricane in last 5–10 years	28.84**	21.69*	21.19*	22.90
	(11.97)	(11.82)	(11.94)	(15.42)
Baseline Mean of Dep Var.	376.05	376.05	376.05	376.05
Dependent Variable: *Violent Crime Rate*				
Minor Hurricane in last 1–4 years	-0.888	0.0305	0.0668	0.0668
	(0.97)	(0.97)	(0.97)	(0.97)
Minor Hurricane in last 5–10 years	-0.489	0.00588	0.0369	0.0369
	(1.03)	(1.00)	(1.00)	(1.00)
Major Hurricane in last 1–4 years	1.489	1.269	1.271	1.477
	(3.00)	(3.27)	(3.27)	(4.05)
Major Hurricane in last 5–10 years	1.864	1.124	1.127	1.319
	(2.68)	(2.75)	(2.75)	(3.42)
Baseline Mean of Dep Var.	42.77	42.77	42.77	42.77
Observations	28249	28249	28249	28249

Note: Col (1): BL TWFE. Col (2): TWFE w/ 1st events only. Col (3): (Sun and Abraham 2021) estimator w/ NT as controls. Col (4): (Sun and Abraham 2021) w/ LT cohort as controls.

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Controlling for multiple hypothesis testing

	Property Crime Rate		Violent Crime Rate	
	(1)	(2)	(3)	(4)
Minor Hurricane in last 1–4 years	-1.214	-1.214	-0.888	-0.888
	(4.17)	(1.030)	(0.97)	(.)
Minor Hurricane in last 5–10 years	-6.306	-6.306	-0.489	-0.489
	(4.25)	(3.851)	(1.03)	(.)
Major Hurricane in last 1–4 years	34.36***	34.356**	1.489	1.489
	(11.66)	(13.946)	(3.00)	(.)
Major Hurricane in last 5–10 years	28.84**	28.842**	1.864	1.864
	(11.97)	(12.839)	(2.68)	(.)
Baseline Mean of Dep Var.	376.05	376.05	42.77	42.77
Observations	28249	28249	28249	28249

Note: Cols (1) and (3) present the baseline two-way fixed-effects (TWFE) estimates for property and violent crime rates, respectively. Cols (2) and (4) report the same estimates with p-values adjusted for multiple hypothesis testing using the sharpened q-value procedure of (Anderson 2008), as reflected in the revised SEs. Because the confidence interval for violent crime rates is extremely wide, the (Anderson 2008) procedure is unable to generate a meaningful SE for that outcome.

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Sensitivity to the choice of different samples

	(1)	(2)	(3)
Dependent Variable: *Property Crime Rate*			
Minor Hurricane in last 1–4 years	-1.214	-3.398	-1.022
	(4.17)	(3.47)	(4.15)
Minor Hurricane in last 5–10 years	-6.306	-4.434	-4.547
	(4.25)	(3.50)	(4.56)
Major Hurricane in last 1–4 years	34.36***	21.05**	30.90***
	(11.66)	(8.66)	(11.46)
Major Hurricane in last 5–10 years	28.84**	19.18**	29.68**
	(11.97)	(8.26)	(13.26)
Baseline Mean of Dep Var.	376.05	374.27	376.05
Dependent Variable: *Violent Crime Rate*			
Minor Hurricane in last 1–4 years	-0.888	-1.502	-0.863
	(0.97)	(0.93)	(0.92)
Minor Hurricane in last 5–10 years	-0.489	-0.502	0.0394
	(1.03)	(0.90)	(1.15)
Major Hurricane in last 1–4 years	1.489	3.039	0.312
	(3.00)	(2.28)	(3.24)
Major Hurricane in last 5–10 years	1.864	1.934	0.903
	(2.68)	(2.01)	(2.40)
Baseline Mean of Dep Var.	42.77	42.5	42.76
Observations	28249	28249	28249

Note: Col (1): Baseline balanced panel. Col (2): Unbalanced panel with hurricane states. Col (3): Balanced panel with hurricane and non-hurricane states.

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Sensitivity to different sample quality checks

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: *Property Crime Rate*					
Minor Hurricane in last 1–4 years	-1.214	-3.089	-1.653	-0.883	1.774
	(4.17)	(4.45)	(4.13)	(4.24)	(4.36)
Minor Hurricane in last 5–10 years	-6.306	-7.164*	-6.639	-6.080	-1.073
	(4.25)	(4.25)	(4.25)	(4.37)	(4.28)
Major Hurricane in last 1–4 years	34.36***	36.60***	34.78***	27.42**	38.16***
	(11.66)	(11.72)	(11.68)	(10.63)	(12.40)
Major Hurricane in last 5–10 years	28.84**	31.04***	29.36**	25.20**	26.86**
	(11.97)	(11.34)	(12.09)	(11.97)	(13.12)
Baseline Mean of Dep Var.	376.05	376.05	376.05	386.46	386.72
Dependent Variable: *Violent Crime Rate*					
Minor Hurricane in last 1–4 years	-0.888	-1.114	-0.902	-0.957	0.697
	(0.97)	(1.04)	(0.98)	(0.98)	(0.93)
Minor Hurricane in last 5–10 years	-0.489	-0.550	-0.507	-0.400	1.158
	(1.03)	(1.04)	(1.05)	(1.04)	(1.04)
Major Hurricane in last 1–4 years	1.489	1.949	1.654	-0.141	3.052
	(3.00)	(2.99)	(3.10)	(2.81)	(2.97)
Major Hurricane in last 5–10 years	1.864	1.850	2.033	0.504	1.072
	(2.68)	(2.59)	(2.77)	(2.50)	(2.84)
Baseline Mean of Dep Var.	42.7	42.7	42.7	43.78	43.66
Observations	28249	26871	27560	25461	24026

Note: Col (1): BL spec. Col (2): Excl. '04–'05. Col (3): Excl 2020.
Col (4): Excl. extreme-crime value counties. Col (5): Excl. counties with ≤ 3 mo/yr crime data.

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